FIELD REFERENCE · LEGO TECHNIC

Technic Gear Atlas

Every gear, bevel, worm, rack, turntable and sprocket in the Technic system — teeth counts, families, and exactly what meshes with what.

SYSTEM	PARTS CATALOGED	SCALE	SHEET
LEGO Technic	21	N.T.S.	1 of 1 · 2026

- Spur — Group A
- Spur — Group B (double-bevel)
- Worm & Crown
- Differential
- Turntables
- Racks
- Isolated / self-mesh only
- Sprockets & chain

Spur — Group A 4 parts

The original Technic pitch (1977–). 8/16/24/40-tooth gears mesh freely with each other and with a matching pinion rack.

8-Tooth Gear

8t · 1977

The smallest common spur gear.

• 16-Tooth Gear • 24-Tooth Gear • 40-Tooth Gear ▬ Gear Rack • 24-Tooth Crown Gear

16-Tooth Gear

16t · 1979

• 8-Tooth Gear • 24-Tooth Gear • 40-Tooth Gear ▬ Gear Rack • 24-Tooth Crown Gear

24-Tooth Gear

24t · 1977

Also sold as a white clutch gear that slips above a torque threshold — meshes identically.

• 8-Tooth Gear • 16-Tooth Gear • 40-Tooth Gear ▬ Gear Rack ⟂ Worm Gear

40-Tooth Gear

40t · 1977

The largest Group A spur gear; a heavy reduction stage.

• 8-Tooth Gear • 16-Tooth Gear • 24-Tooth Gear ▬ Gear Rack ⟂ Worm Gear

- • 24-Tooth Crown Gear

- • 24-Tooth Crown Gear

Spur — Group B (double-bevel) 3 parts

A finer, later pitch (1999–). 12/20/36-tooth double-bevel gears mesh with each other and a matching rack — but not with Group A.

12-Tooth Double-Bevel

12t · 1999

Runs parallel like a spur gear, or at 90° against another bevel.

- ⟂ 24-Tooth Crown Gear
- ⟂ 20-Tooth Double-Bevel
- • 36-Tooth Double-Bevel
- ▬ Gear Rack
- ⟂ Differential Housing

20-Tooth Double-Bevel

20t · 1999

- ⟂ 12-Tooth Double-Bevel
- • 36-Tooth Double-Bevel
- ▬ Gear Rack

36-Tooth Double-Bevel

36t · 2002

The largest System-pitch gear; treated as parallel-only here.

- • 12-Tooth Double-Bevel
- • 20-Tooth Double-Bevel
- ▬ Gear Rack

Worm & Crown 2 parts

Special-purpose Group A partners: a self-locking worm screw and a crown gear that bridges parallel and right-angle drives.

Worm Gear (Screw)

1 thread · 1985

Non-backdrivable — locks the output. Turns only 24t/40t Group A gears.

- ⟂ 24-Tooth Gear
- ⟂ 40-Tooth Gear
- ⟂ 24-Tooth Crown Gear

24-Tooth Crown Gear

24t · 1977

Doubles as a Group A spur gear and mates 90° with bevels.

- • 8-Tooth Gear
- • 16-Tooth Gear
- • 24-Tooth Gear
- • 40-Tooth Gear
- ▬ Gear Rack
- ⟂ Worm Gear
- ⟂ 12-Tooth Double-Bevel

⟂ 14-Tooth Bevel

Differential `1 part`

Splits torque between two outputs through an internal bevel gear set housed in a ring.

Differential Housing

~28t ring

Its input ring meshes at 90° with a bevel pinion.

⟂ 12-Tooth Double-Bevel ⟂ 14-Tooth Bevel

Turntables `2 parts`

Large slewing rings for rotating platforms — driven at their perimeter, not charted against pinion gears here.

Small Turntable

28t · 2012

No standard direct mesh

Large Turntable

56t · 1990

Studless since 2004; a 60-tooth bevel-edge variant ran 2015–2022.

No standard direct mesh

Racks `2 parts`

Straight or curved linear tracks. Come in two pitches, one per spur family.

Gear Rack — Group A pitch

linear

Matches any 8/16/24/40-tooth pinion.

▬ 8-Tooth Gear ▬ 16-Tooth Gear ▬ 24-Tooth Gear ▬ 40-Tooth Gear ▬ 24-Tooth Crown Gear

Gear Rack — System pitch

linear

Matches 12/20/36-tooth double-bevel pinions.

▬ 12-Tooth Double-Bevel ▬ 20-Tooth Double-Bevel ▬ 36-Tooth Double-Bevel

Isolated / self-mesh only `3 parts`

Bevels and wheels deliberately built to mesh only with an identical copy of themselves.

14-Tooth Bevel (original)

14t · 1980

Superseded by the 12t bevel; meshes only with its own kind and the differential.

⊥ **24-Tooth Crown Gear** ⊥ **Differential Housing**

22t / 14t Thick Bevel

22t / 14t · 2022

A newer, stronger pair deliberately isolated from every other bevel.

No standard direct mesh

Knob Wheel

6 lobes

Chunky positive-drive wheel; meshes only with another knob wheel.

No standard direct mesh

Sprockets & chain `4 parts`

A separate drive system entirely — chain-tread links ride on sprockets and never mesh tooth-to-tooth with a gear.

Small Sprocket

6t · 57520

Chain-drive only.

No standard direct mesh

Large Sprocket

10t · 57519

No standard direct mesh

XL Sprocket

14t · 42529

No standard direct mesh

Thin/Dual Sprocket

20t · 32089

One chain loop can run mixed sprocket sizes together.

No standard direct mesh

Compatibility matrix

Standard direct interfaces only — same-axle-spacing meshes a builder gets by just putting the parts together. Community "offset spacing" tricks that force otherwise-incompatible pitches to mesh aren't charted here; see the notes below.

● parallel / spur mesh ⊥ right-angle mesh (bevel · crown · worm · differential) ▬ rack & pinion (linear)

· no standard mesh

	8-Tooth Gear	16-Tooth Gear	24-Tooth Gear	40-Tooth Gear	12-Tooth Double-Bevel	20-Tooth Double-Bevel	36-Tooth Double-Bevel	Worm Gear (Screw)	24-Tooth Crown Gear	Differential Housing	Small Turntable	Large Turntable	Gear Rack – Group A pitch	Gear Rack – System pitch	14-Tooth Bevel (original)	22t / 14t Thick Bevel	Knob Wheel	Small Sprocket
8-Tooth Gear		●	●	●	·	·	·	·	●	·	·	·	▬	·	·	·	·	·
16-Tooth Gear	●		●	●	·	·	·	·	●	·	·	·	▬	·	·	·	·	·
24-Tooth Gear	●	●		●	·	·	·	⊥	●	·	·	·	▬	·	·	·	·	·
40-Tooth Gear	●	●	●		·	·	·	⊥	●	·	·	·	▬	·	·	·	·	·
12-Tooth Double-Bevel	·	·	·	·		⊥	●	·	⊥	⊥	·	·	·	▬	·	·	·	·
20-Tooth Double-Bevel	·	·	·	·	⊥		●	·	·	·	·	·	·	▬	·	·	·	·
36-Tooth Double-Bevel	·	·	·	·	●	●		·	·	·	·	·	·	▬	·	·	·	·
Worm Gear (Screw)	·	·	⊥	⊥	·	·	·		⊥	·	·	·	·	·	·	·	·	·
24-Tooth Crown Gear	●	●	●	●	⊥	·	·	⊥		·	·	·	▬	·	⊥	·	·	·
Differential Housing	·	·	·	·	⊥	·	·	·	·		·	·	·	·	⊥	·	·	·
Small Turntable	·	·	·	·	·	·	·	·	·	·		·	·	·	·	·	·	·
Large Turntable	·	·	·	·	·	·	·	·	·	·	·		·	·	·	·	·	·
Gear Rack – Group A pitch	▬	▬	▬	▬	·	·	·	·	▬	·	·	·		·	·	·	·	·
Gear Rack – System pitch	·	·	·	·	▬	▬	▬	·	·	·	·	·	·		·	·	·	·
14-Tooth Bevel (original)	·	·	·	·	·	·	·	·	⊥	⊥	·	·	·	·		·	·	·
22t / 14t Thick Bevel	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·		·	·
Knob Wheel	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·		·
Small Sprocket	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	
Large Sprocket	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·
XL Sprocket	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·
Thin/Dual Sprocket	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·

Scroll to see every part × part relationship. Row and column order matches the family sections above

Notes & caveats

- **Group A vs. Group B don't natively mesh.** The 8/16/24/40-tooth family and the 12/20/36-tooth double-bevel family use different tooth pitches. Builders can sometimes force a mesh (e.g. 24t↔12t) by spacing axles off-grid — a known community technique, not a designed interface, so it isn't charted here.
- **The worm gear is one-way.** It can turn a 24t or 40t gear, but a 24t/40t gear can't turn it back — useful for locking mechanisms and unsuitable anywhere back-drive is needed.
- **Turntables are driven at the rim**, typically by a small gear or worm pressed against their outer teeth, but which pinion pitch is 'standard' isn't consistently documented — left off the matrix rather than guessed.
- **Sprockets never mesh with gears** at all. They connect only through Technic chain-tread links, and a single chain loop happily mixes sprocket sizes (e.g. a 6t driving a 14t) — that's the one compatibility axis in this subsystem, and it's a chain relationship, not a tooth mesh, so it isn't in the matrix.
- Sourced from fan references (Rebrickable, BrickNerd, Technicopedia and community write-ups), not LEGO's own documentation — solid for planning a build, not spec-grade for manufacturing.
